

# FINITE SOLUBLE GROUPS NEED NOT ADMIT INCREASING UNREFINABLE SUBGROUP CHAINS

RICHIE SATER 

**ABSTRACT.** An unrefinable subgroup chain is called increasing when its successive indices are non-decreasing. We show that not every finite group has such a chain, even among soluble groups. For every odd prime  $p$ , we construct a soluble matrix group  $G_p \leq \text{GL}_5(p)$  of order  $2^6 p^8$  with no increasing unrefinable chain. The mechanism is a nested maximal-index-spectrum trap: every candidate chain is forced through two steps of index  $p^2$  to a subgroup  $N$ , where the next admissible index is 2. All remaining lower indices would then be 2, contradicting  $p \mid |N|$ . The construction adapts module layers from an example of Kohler. An ordinary group-theoretic argument proves the result for all odd  $p$ ; GAP calculations at  $p = 3, 5, 7$  corroborate selected instances.

## 1. INTRODUCTION

Let  $G$  be a nontrivial finite group. An unrefinable chain of subgroups

$$1 = G_0 < G_1 < \cdots < G_n = G$$

is one in which  $G_{i-1}$  is maximal in  $G_i$  for every  $i$ . Associate with this chain the indices  $j_i = |G_i : G_{i-1}|$ . We call the chain *increasing* if

$$j_1 \leq j_2 \leq \cdots \leq j_n.$$

Monakhov and Sokhor call such a chain a  $<$ -chain [3]. More precisely, they prove that a finite group is supersoluble if it has two maximal chains of the same length, one a  $<$ -chain and the other a  $>$ -chain, and they classify the finite groups for which every maximal chain in every proper subgroup is a  $<$ -chain or a  $>$ -chain. In correspondence beginning on August 8, 2026, they communicated to the author the following question, stated here in equivalent terminology.<sup>1</sup>

Does every finite group contain an increasing unrefinable subgroup chain?

Kuperberg and Zieve study a different Jordan-type property for subgroup-lattice intervals: under additional hypotheses, all maximal chains between a point stabilizer and a transitive permutation group have the same relative indices up to permutation [2]. Their invariance question across all maximal chains is distinct from the present existential ordering question, which asks whether some full chain has nondecreasing indices.

The condition is existential, so exhibiting a few decreasing chains cannot answer the question. A counterexample must control every possible maximal step without requiring a description of the entire subgroup lattice. We do this by controlling the maximal-index spectra of three nested levels.

**Theorem 1.** *For every odd prime  $p$ , there is a soluble group  $G_p \leq \text{GL}_5(p)$  of order  $2^6 p^8$  which has no increasing unrefinable subgroup chain.*

---

*Date:* August 29, 2026.

*2020 Mathematics Subject Classification.* Primary 20D30; Secondary 20D10, 20F22.

*Key words and phrases.* finite groups, maximal chains, maximal-subgroup indices, soluble groups, semidirect products.

<sup>1</sup>The correspondence is the source through which the author learned of the question; no claim is made here about its original formulation. Reference [3] supplies terminology and background rather than a published source of this universal-existence question.

The proof is organized around the following downward pattern, where the arrows are maximal-subgroup steps and their labels are the corresponding indices:

$$(1) \quad \mathcal{I}(G_p) = \{2, p^2\} \xrightarrow{p^2} \mathcal{I}(M_X) = \mathcal{I}(M_Y) = \{2, p^2, p^4\} \xrightarrow{p^2} \mathcal{I}(N) = \{2, p^4\}.$$

Reading downward, monotonicity forces a candidate chain through the two index- $p^2$  steps to  $N$ . There the only admissible next index is 2, so every remaining lower index would also be 2. This would make  $N$  a 2-group, contradicting  $p \mid |N|$ . Proposition 2 isolates this argument from the group construction.

The groups themselves adapt the  $t = 2$  case of Kohler's construction in Section 4, especially Theorem 4.1, of [1]. Kohler used its module structure to show that restrictions on maximal-subgroup indices need not pass to subgroups. Here two nonisomorphic irreducible layers of dimension 2 and an irreducible commutator layer of dimension 4 are used for a different purpose: they realize the three spectra in (1). Thus the reusable feature is not the matrix presentation by itself, but the way successive module quotients engineer a prescribed path of maximal indices.

The smallest displayed member is  $G_3$ , of order  $2^6 3^8 = 419904$ ; no global minimality is claimed. The proof of Theorem 1 is entirely ordinary mathematics and uses no classification. The repository's exact GAP computations check the predicted spectra for  $p = 3, 5, 7$  and exhaustively search the case  $p = 3$ ; they are finite corroboration rather than a proof of the universal statement. Section 2 gives the abstract obstruction, Section 3 constructs the required module layers, and Section 4 computes their spectra and proves the theorem. Further questions are recorded in Section 5, while the explicit matrices and computational boundary appear in Appendix A.

## 2. A NESTED INDEX-SPECTRUM TRAP

For a finite group  $A$ , write

$$\mathcal{I}(A) = \{|A : B| \mid B \text{ is maximal in } A\}.$$

The next proposition is the conceptual core of the proof. It replaces a global search through the subgroup lattice by three local spectra and two incidence statements.

**Proposition 2** (Nested spectrum trap). *Let  $p$  be an odd prime, and let  $G$  be a finite group with subgroups  $M_1, M_2, N$  such that  $M_1$  and  $M_2$  are maximal in  $G$ , while  $N$  is maximal in each  $M_i$ . Suppose that*

$$\begin{aligned} \mathcal{I}(G) &= \{2, p^2\}, \\ \mathcal{I}(M_1) &= \mathcal{I}(M_2) = \{2, p^2, p^4\}, \\ \mathcal{I}(N) &= \{2, p^4\}. \end{aligned}$$

*Assume also that*

- (1) *every maximal subgroup of  $G$  of index  $p^2$  is  $G$ -conjugate to  $M_1$  or  $M_2$ ;*
- (2) *for  $i = 1, 2$ , every maximal subgroup of  $M_i$  of index  $p^2$  is  $M_i$ -conjugate to  $N$ ; and*
- (3)  *$p$  divides  $|N|$ .*

*Then  $G$  has no increasing unrefinable subgroup chain.*

*Proof.* Suppose that

$$1 = G_0 < G_1 < \cdots < G_n = G$$

is unrefinable and that  $j_i = |G_i : G_{i-1}|$  is nondecreasing. The last index belongs to  $\mathcal{I}(G)$ . If  $j_n = 2$ , then every  $j_i$  equals 2, so  $|G|$  is a power of 2; this contradicts  $p \mid |N|$  and  $N \leq G$ . Hence  $j_n = p^2$ . After conjugating the chain in  $G$ , we may take  $G_{n-1} = M_i$  for some  $i \in \{1, 2\}$ .

Now  $j_{n-1} \leq p^2$ . The spectrum of  $M_i$  leaves only 2 and  $p^2$ . The value 2 would again force every preceding index to be 2, contrary to  $p \mid |N|$  and  $N \leq M_i$ . Thus  $j_{n-1} = p^2$ . Conjugating the lower part of the chain inside  $M_i$ , we may take  $G_{n-2} = N$ .

Finally,  $j_{n-2} \leq p^2$ , whereas  $\mathcal{I}(N) = \{2, p^4\}$ . Therefore  $j_{n-2} = 2$ , and monotonicity forces all earlier indices to equal 2. This would make  $|N|$  a power of 2, contradicting  $p \mid |N|$ .  $\square$

The proof uses only the labeled local data

$$\{2, p^2\} \longrightarrow \{2, p^2, p^4\} \longrightarrow \{2, p^4\}.$$

In particular, one need not enumerate maximal chains or even determine the full subgroup lattice. The remaining task is to realize this diagram in a soluble group.

### 3. THE GROUP FAMILY AND ITS MODULE LAYERS

We first record the standard maximal-subgroup fact that converts irreducible module dimensions into subgroup indices. We then construct three layers of dimensions 2, 2, 4, exactly the dimensions required by Proposition 2.

#### 3.1. Maximal subgroups of a coprime semidirect product.

**Lemma 3.** *Let  $V$  be an elementary abelian  $p$ -group, let  $H$  be a  $p'$ -group acting on  $V$ , put  $K = V \rtimes H$ , and let  $\pi : K \rightarrow H$  be the natural projection. Every maximal subgroup of  $K$  has one of the following forms:*

- (1)  $\pi^{-1}(J) = V \rtimes J$ , where  $J$  is maximal in  $H$ ;
- (2) a  $V$ -conjugate of  $W \rtimes H$ , where  $W$  is a maximal proper  $H$ -submodule of  $V$ .

*Proof.* Let  $T$  be maximal in  $K$ . If  $V \leq T$ , then  $T/V$  is maximal in  $K/V \cong H$ , which gives the first form. Suppose  $V \not\leq T$ . Then  $VT = K$ . Put  $W = V \cap T$ . The group  $T$  normalizes  $W$ , while  $V$  centralizes  $W$ ; hence  $W \triangleleft K$ . In  $K/W$ , both  $T/W$  and  $WH/W$  are complements to  $V/W$ . By the Schur-Zassenhaus theorem, these complements are conjugate by an element of  $V/W$ . Lifting this conjugacy shows that  $T$  is  $V$ -conjugate to  $W \rtimes H$ . Moreover, maximality of  $T$  is equivalent to maximality of  $W$  among the proper  $H$ -submodules of  $V$ .  $\square$

If  $H$  is a 2-group, the subgroups of the first type have index 2. If  $V$  is completely reducible, the indices of the second type are the orders of the irreducible quotients of  $V$ . This is the interface used below: an irreducible quotient of dimension  $d$  over  $\mathbb{F}_p$  produces index  $p^d$ .

**3.2. The matrix group.** Fix an odd prime  $p$  and put  $F = \mathbb{F}_p$ . Let

$$s = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad t = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad D = \langle s, t \rangle \leq \mathrm{GL}_2(F).$$

Then  $D \cong D_8$ , where  $D_8$  has order 8. The natural  $FD$ -module  $U = F^2$  is absolutely irreducible. Indeed, after extending scalars to an algebraic closure, the only  $t$ -invariant lines are the two coordinate lines, and  $s$  interchanges them.

Let  $H = D \times D$ . The  $p$ -group on which  $H$  will act is

$$L = \left\{ \ell(x, y, z) = \begin{pmatrix} I_2 & x & z \\ 0 & 1 & y \\ 0 & 0 & I_2 \end{pmatrix} : x \in F^{2 \times 1}, y \in F^{1 \times 2}, z \in F^{2 \times 2} \right\}.$$

Here the block sizes are 2, 1, 2. The following two calculations identify the degree-one coordinates and their commutator layer. Direct multiplication gives

$$(2) \quad \ell(x, y, z)\ell(x', y', z') = \ell(x + x', y + y', z + z' + xy'),$$

and, with  $[a, b] = a^{-1}b^{-1}ab$ ,

$$(3) \quad [\ell(x, y, z), \ell(x', y', z')] = \ell(0, 0, xy' - x'y).$$

Thus  $|L| = p^8$ . Put

$$Z = \{\ell(0, 0, z)\}, \quad X = \{\ell(x, 0, 0)\}, \quad Y = \{\ell(0, y, 0)\}.$$

The group  $Z$  is central in  $L$ . The outer products  $xy$  span  $F^{2 \times 2}$ , so (3) gives  $L' = Z$ . Since  $p$  is odd,  $L$  has exponent  $p$ ; explicitly,

$$\ell(x, y, z)^p = \ell\left(0, 0, pz + \binom{p}{2}xy\right) = 1.$$

The standard formula for the Frattini subgroup of a finite  $p$ -group now gives

$$(4) \quad \Phi(L) = L^p L' = Z.$$

The four coordinates in  $x$  and  $y$  and the four independent coordinates in  $z$  also make the order  $p^{4+4} = p^8$  transparent.

Embed  $H$  in  $\text{GL}_5(F)$  by identifying  $(A, B) \in D \times D$  with  $\text{diag}(A, 1, B)$ . Conjugation acts by

$$x \mapsto Ax, \quad y \mapsto yB^{-1}, \quad z \mapsto AzB^{-1}.$$

Consequently,  $H$  normalizes both  $L$  and  $Z$ . Define

$$G_p = L \rtimes H \leq \text{GL}_5(p).$$

Then  $Z \triangleleft G_p$  and  $|G_p| = p^8 \cdot 8^2 = 2^6 p^8$ . Since both  $L$  and  $H$  are soluble, so is  $G_p$ .

**3.3. The layers of dimensions 2, 2, 4.** Set  $\overline{X} = XZ/Z$  and  $\overline{Y} = YZ/Z$ . As an  $H$ -module,

$$(5) \quad L/Z = \overline{X} \oplus \overline{Y},$$

where  $\overline{X}$  and  $\overline{Y}$  are nonisomorphic irreducible modules of dimension 2. The first factor of  $D \times D$  acts naturally on  $\overline{X}$  and the second acts trivially; the second factor acts dually on  $\overline{Y}$  and the first acts trivially. In particular,

$$C_H(\overline{X}) = 1 \times D, \quad C_H(\overline{Y}) = D \times 1.$$

The distinct action kernels show that the modules are nonisomorphic. Since  $p \nmid |H| = 64$ , Maschke's theorem applies, and the only maximal  $H$ -submodules of  $L/Z$  are  $\overline{X}$  and  $\overline{Y}$ .

The commutator layer is

$$(6) \quad Z \cong U \boxtimes U^*,$$

the external tensor product for the two direct factors of  $H$ . Hence  $Z$  is an irreducible  $H$ -module of dimension 4. To see this over the given field, extend scalars to an algebraic closure  $\overline{F}$ : both  $U \otimes_F \overline{F}$  and  $U^* \otimes_F \overline{F}$  remain irreducible, and their external tensor product is irreducible for  $D \times D$ .

The role of these dimensions can now be read backward from the desired trap. The two dimension-2 quotients in  $L/Z$  create the two top steps of index  $p^2$ . After one such quotient is removed, the remaining elementary abelian normal subgroup has irreducible layers of dimensions 2 and 4. Removing the second dimension-2 layer leaves the irreducible module  $Z$ , whose only non-2 maximal index is  $p^4$ . This is precisely the transition displayed in (1).

## 4. REALIZING THE SPECTRUM TRAP

We now compute only the maximal-subgroup data required by Proposition 2. The calculation has three levels: the quotient  $G_p/Z$ , the two middle semidirect products, and the terminal group  $Z \rtimes H$ .

**Lemma 4.** *Put*

$$M_X = (Z \times X) \rtimes H, \quad M_Y = (Z \times Y) \rtimes H, \quad N = Z \rtimes H.$$

*The subgroups  $M_X$  and  $M_Y$  are maximal in  $G_p$  of index  $p^2$ , and  $N$  is maximal in each of them with index  $p^2$ . Moreover,*

$$\begin{aligned} \mathcal{I}(G_p) &= \{2, p^2\}, \\ \mathcal{I}(M_X) &= \mathcal{I}(M_Y) = \{2, p^2, p^4\}, \\ \mathcal{I}(N) &= \{2, p^4\}. \end{aligned}$$

*Every maximal subgroup of  $G_p$  of index  $p^2$  is  $G_p$ -conjugate to  $M_X$  or  $M_Y$ . Every maximal subgroup of  $M_X$  of index  $p^2$  is  $M_X$ -conjugate to  $N$ , and every maximal subgroup of  $M_Y$  of index  $p^2$  is  $M_Y$ -conjugate to  $N$ .*

*Proof. The top level.* First, every maximal subgroup  $M$  of  $G_p$  contains  $Z$ . If  $Z \not\leq M$ , then maximality gives  $G_p = ZM$ , and the modular law together with (4) gives

$$L = L \cap ZM = Z(L \cap M) = \Phi(L)(L \cap M).$$

Recall that  $L = \Phi(L)K$  for a subgroup  $K \leq L$  implies  $K = L$ : otherwise, a maximal subgroup of  $L$  containing  $K$  would contain both  $K$  and  $\Phi(L)$ , a contradiction. Applying this fact to  $K = L \cap M$  forces  $L \cap M = L$ , so  $Z \leq M$ , again a contradiction.

It follows that the maximal subgroups of  $G_p$  are the inverse images of the maximal subgroups of

$$G_p/Z = (\bar{X} \oplus \bar{Y}) \rtimes H.$$

Lemma 3 and (5) give index 2 from maximal subgroups of the 2-group  $H$ , and index  $p^2$  from either irreducible quotient. They also show that the index- $p^2$  maximal subgroups are conjugate to  $M_X$  or  $M_Y$ . This proves the top spectrum and its incidence statement.

*The middle level.* In  $M_X$ , the normal elementary abelian subgroup  $Z \times X$  is the direct sum of nonisomorphic irreducible  $H$ -modules of dimensions 4 and 2. Its only maximal  $H$ -submodules are therefore  $Z$  and  $X$ . Lemma 3 gives the indices  $2, p^2, p^4$ ; the index- $p^2$  case retains  $Z$  and hence gives an  $M_X$ -conjugate of  $N$ . The same argument applies to  $M_Y$ .

*The terminal level.* The  $H$ -module  $Z$  is irreducible of dimension 4. Applying Lemma 3 to  $N = Z \rtimes H$  gives  $\mathcal{I}(N) = \{2, p^4\}$ . These three calculations establish every spectrum and incidence needed for the trap.  $\square$

*Proof of Theorem 1.* The construction gives a soluble group  $G_p$  of order  $2^6 p^8$ . Lemma 4 supplies the three spectra and both incidence statements in Proposition 2; moreover,  $p$  divides  $|N| = 2^6 p^4$ . The proposition therefore excludes every increasing unrefinable subgroup chain in  $G_p$ .  $\square$

The proof has thus separated the construction from the obstruction. Module theory establishes the local diagram, and Proposition 2 turns that diagram into the global nonexistence result.

## 5. CONCLUSION

The spectrum trap is a small directed system rather than a subgroup-lattice classification: its vertices record maximal-index spectra, and its marked arrows record which maximal subgroups can follow an admissible index. This viewpoint may be useful whenever one wants to force prescribed

risers or drops in unrefinable chains. Different group constructions could realize the same local diagram.

In this example, the two nonisomorphic dimension-2 layers and the irreducible dimension-4 layer are structurally essential to the displayed pattern. Coprime action supplies complete reducibility, while the 2-group complement makes every quotient-side maximal index equal to 2. Neither a classification theorem nor a computation enters the uniform proof.

**Further questions.** Three natural questions remain. What is the least order of a counterexample? Which labeled spectrum diagrams can be realized by finite soluble groups? And can similar module-layer constructions prescribe more complicated local patterns in maximal chains? The present argument neither establishes the minimality of  $G_3$  nor classifies groups that admit increasing chains.

#### APPENDIX A. THE EXPLICIT GROUP $G_3$ AND THE COMPUTATIONAL BOUNDARY

This appendix gives one concrete representative and states exactly what the repository computations check. It is not needed for the proof for arbitrary odd  $p$ .

All matrices below are over  $\mathbb{F}_3$ , so  $-1 = 2$ . Let  $E_{ij}$  denote the standard matrix unit, write  $e_{ij} = I_5 + E_{ij}$ , and set

$$\begin{aligned} a_s &= \text{diag}(s, 1, I_2), & a_t &= \text{diag}(t, 1, I_2), \\ b_s &= \text{diag}(I_2, 1, s), & b_t &= \text{diag}(I_2, 1, t). \end{aligned}$$

Then the smallest displayed example is

$$(7) \quad G_3 = \langle e_{13}, e_{23}, e_{34}, e_{35}, a_s, a_t, b_s, b_t \rangle \leq \text{GL}_5(3).$$

Define

$$\begin{aligned} Z_3 &= \langle e_{14}, e_{15}, e_{24}, e_{25} \rangle, \\ H_3 &= \langle a_s, a_t, b_s, b_t \rangle, \\ M_{X,3} &= \langle Z_3, e_{13}, e_{23}, H_3 \rangle, \\ M_{Y,3} &= \langle Z_3, e_{34}, e_{35}, H_3 \rangle, & N_3 &= \langle Z_3, H_3 \rangle. \end{aligned}$$

Here  $|H_3| = 64$ . The relevant orders and complete maximal-index spectra are

| $A$                | $ A $  | $\mathcal{I}(A)$ |
|--------------------|--------|------------------|
| $G_3$              | 419904 | $\{2, 9\}$       |
| $M_{X,3}, M_{Y,3}$ | 46656  | $\{2, 9, 81\}$   |
| $N_3$              | 5184   | $\{2, 81\}$ .    |

The ordinary argument in Sections 2–4 proves Theorem 1 for every odd prime. The standalone file `G3-verification.g` separately defines the eight generators in (7) as literal  $5 \times 5$  matrices. Using standard GAP commands and no optional package, it checks the named orders and maximal inclusions, enumerates every conjugacy class of maximal subgroups in the four relevant groups, checks coverage of the marked conjugacy classes, and performs an exhaustive top-down search for an increasing chain in  $G_3$ . Captured successful runs use GAP 4.11.1 and GAP 4.16.0.

The family tests in `tests/counterexample.g` instantiate  $p = 3, 5, 7$  and reproduce the predicted spectra and incidences. The separate `tests/mmc-crosscheck.g` compares the repository's chain recurrence with exhaustive index-sequence enumeration for all 144 groups of order at most 32. This last test checks the implementation of the recurrence, not the theorem's general construction. Together these exact computations can catch transcription and implementation errors in the displayed instances; they do not replace the uniform proof.

The theorem-related GAP scripts and captured outputs described above are archived in release v1.1.0 of the accompanying repository [4].

## ACKNOWLEDGMENTS

The author thanks V. S. Monakhov and I. L. Sokhor for bringing the question to his attention through their correspondence.

## CODE AND DATA AVAILABILITY

The computational supplement is publicly available in the verification repository. Release **v1.1.0** contains the standalone verifier and captured outputs from GAP 4.11.1 and GAP 4.16.0. Source code is licensed under the MIT License, and the manuscript and documentation are licensed under CC BY 4.0.

## GENERATIVE-AI DISCLOSURE

From 8 to 29 August 2026, the author used the OpenAI Codex CLI, versions 0.147.0 and 0.150.1 with model GPT-5.6-sol, through a locally installed command-line client for mathematical exploration, literature searches, GAP and regression-test development, proof auditing, and editorial revision; the theorem is proved by the mathematical argument presented here, the reported computations have the stated finite scope, and the author takes responsibility for the manuscript.

## REFERENCES

- [1] J. Kohler, *Finite groups with all maximal subgroups of prime or prime square index*, Canad. J. Math. **16** (1964), 435–442. doi:10.4153/CJM-1964-046-6.
- [2] G. Kuperberg and M. E. Zieve, *Analogues of the Jordan–Hölder theorem for transitive  $G$ -sets*, arXiv:0712.4142 (2007). doi:10.48550/arXiv.0712.4142.
- [3] V. S. Monakhov and I. L. Sokhor, *On indices of maximal chains in finite groups*, Results Math. **80** (2025), article 155. doi:10.1007/s00025-025-02468-5.
- [4] R. Sater, *A negative answer to a question on monotone maximal chains in finite groups*, version 1.1.0 preprint and verification archive (2026), doi:10.5281/zenodo.21878837.

INDEPENDENT RESEARCHER, UNITED STATES

Email address: richiesater@gmail.com